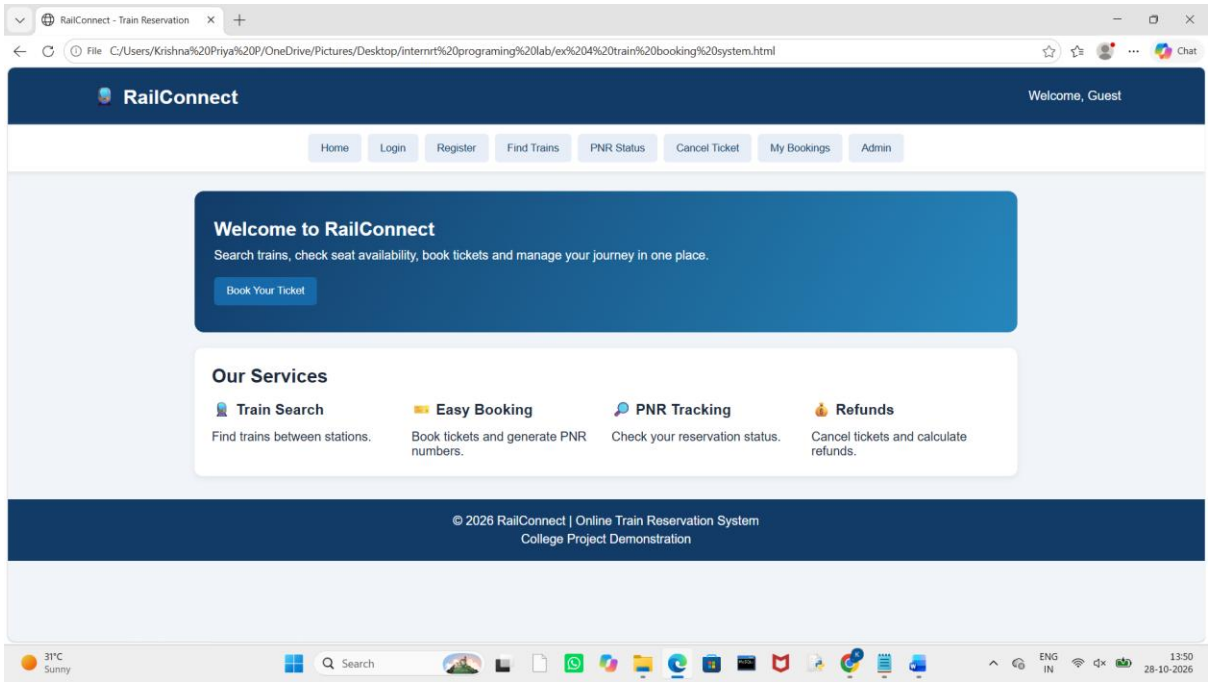



```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>RailConnect - Train Reservation</title>

<style>
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: Arial, sans-serif;
}

body {
  background: #f1f5f9;
  color: #1e293b;
```

```
}

header {
  background: #123b68;
  color: white;
  padding: 20px 7%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
}

header h1 { font-size: 25px; }

nav {
  background: #ffffff;
  padding: 12px;
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
  gap: 8px;
  box-shadow: 0 2px 5px #ddd;
}

button {
  background: #1769aa;
  color: white;
  padding: 10px 16px;
  border: none;
  border-radius: 5px;
```

```
    cursor: pointer;
}

button:hover { background: #0d4778; }

nav button { background: #e7eef7; color: #123b68; }
nav button:hover { background: #cbdcf0; }

.container {
    width: 92%;
    max-width: 1050px;
    margin: 25px auto;
}

.hero {
    background: linear-gradient(120deg, #123b68, #2687bd);
    color: white;
    padding: 35px 25px;
    border-radius: 12px;
    margin-bottom: 20px;
}

.hero h2 { margin-bottom: 10px; }

.panel {
    background: white;
    padding: 22px;
    margin-bottom: 20px;
    border-radius: 10px;
    box-shadow: 0 2px 8px #dce3eb;
}
```

```
h2, h3 { margin-bottom: 15px; }

.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 14px;
}

label {
  display: block;
  margin-bottom: 6px;
  font-weight: bold;
  font-size: 14px;
}

input, select {
  width: 100%;
  padding: 11px;
  border: 1px solid #cbd5e1;
  border-radius: 5px;
  margin-bottom: 12px;
  background: white;
}

.full { grid-column: 1 / -1; }

.train-card {
  border: 1px solid #d5e0ec;
  padding: 17px;
  border-radius: 8px;
```

```
    margin-top: 12px;
}

.train-card h3 { color: #1769aa; }

.train-info {
    display: flex;
    justify-content: space-between;
    flex-wrap: wrap;
    gap: 10px;
    margin: 12px 0;
}

.hidden { display: none !important; }

.message {
    padding: 12px;
    margin: 12px 0;
    background: #e0f2fe;
    color: #075985;
    border-radius: 5px;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 12px;
}

th, td {
    padding: 10px;
```

```
  border: 1px solid #ddd;
  text-align: left;
}

th { background: #eaf1f8; }

.table-wrap { overflow-x: auto; }

footer {
  text-align: center;
  background: #123b68;
  color: white;
  padding: 18px;
  margin-top: 30px;
}

.danger { background: #c62828; }
.green { background: #238636; }

@media(max-width:600px) {
  header { justify-content: center; text-align: center; }
  .hero { padding: 25px 18px; }
  .container { width: 94%; }
  .panel { padding: 15px; }
}
</style>
</head>

<body>

<header>
```

```
  <h1>🚆 RailConnect</h1>
  <span id="userLabel">Welcome, Guest</span>
</header>

<nav>
  <button onclick="showSection('home')">Home</button>
  <button onclick="showSection('login')">Login</button>
  <button onclick="showSection('register')">Register</button>
  <button onclick="showSection('search')">Find Trains</button>
  <button onclick="showSection('pnr')">PNR Status</button>
  <button onclick="showSection('cancel')">Cancel Ticket</button>
  <button onclick="showSection('history')">My Bookings</button>
  <button onclick="showSection('admin')">Admin</button>
</nav>

<div class="container">

<!-- HOME -->
<section id="home">
  <div class="hero">
    <h2>Welcome to RailConnect</h2>
    <p>Search trains, check seat availability, book tickets and manage your journey in one place.</p>
    <br>
    <button onclick="showSection('search')">Book Your Ticket</button>
  </div>

  <div class="panel">
    <h2>Our Services</h2>
    <div class="form-grid">
      <div><h3>🚆 Train Search</h3><p>Find trains between stations.</p></div>
```

```
        <div><h3>🎫 Easy Booking</h3><p>Book tickets and generate PNR numbers.</p></div>
        <div><h3>🔎 PNR Tracking</h3><p>Check your reservation status.</p></div>
        <div><h3>💰 Refunds</h3><p>Cancel tickets and calculate refunds.</p></div>
      </div>
    </div>
</section>

<!-- REGISTER -->
<section id="register" class="panel hidden">
    <h2>Create Account</h2>
    <form onsubmit="registerUser(event)">
      <label>Full Name</label>
      <input id="regName" required>

      <label>Email Address</label>
      <input id="regEmail" type="email" required>

      <label>Password</label>
      <input id="regPassword" type="password" minlength="4" required>

      <button type="submit">Register</button>
    </form>
    <div id="regMsg"></div>
</section>

<!-- LOGIN -->
<section id="login" class="panel hidden">
    <h2>User Login</h2>
    <form onsubmit="loginUser(event)">
      <label>Email</label>
      <input id="loginEmail" type="email" required>
```

```
      <label>Password</label>
      <input id="loginPassword" type="password" required>

      <button type="submit">Login</button>
    </form>
    <p style="margin-top:12px">Demo admin: admin@rail.com / admin123</p>
    <div id="loginMsg"></div>
</section>

<!-- TRAIN SEARCH -->
<section id="search" class="panel hidden">
    <h2>Search Available Trains</h2>
    <form onsubmit="searchTrains(event)">
      <div class="form-grid">
        <div>
          <label>Origin Station</label>
          <select id="origin" required>
            <option value="">Select origin</option>
            <option>Chennai</option>
            <option>Bengaluru</option>
            <option>Coimbatore</option>
            <option>Madurai</option>
            <option>Hyderabad</option>
            <option>Mumbai</option>
          </select>
        </div>

        <div>
          <label>Destination Station</label>
          <select id="destination" required>
```

```
            <option value="">Select destination</option>
            <option>Chennai</option>
            <option>Bengaluru</option>
            <option>Coimbatore</option>
            <option>Madurai</option>
            <option>Hyderabad</option>
            <option>Mumbai</option>
          </select>
        </div>

        <div>
          <label>Journey Date</label>
          <input type="date" id="journeyDate" required>
        </div>

        <div>
          <label>Travel Class</label>
          <select id="travelClass">
            <option value="Sleeper">Sleeper (SL)</option>
            <option value="AC 3 Tier">AC 3 Tier (3A)</option>
            <option value="AC 2 Tier">AC 2 Tier (2A)</option>
            <option value="Chair Car">Chair Car (CC)</option>
          </select>
        </div>
      </div>
      <button type="submit">Search Trains</button>
    </form>

    <div id="trainResults"></div>
  </section>
```

```
<!-- PASSENGER DETAILS -->
<section id="booking" class="panel hidden">
  <h2>Passenger Details</h2>
  <div id="selectedTrain" class="message"></div>

  <form onsubmit="confirmBooking(event)">
    <div class="form-grid">
      <div>
        <label>Passenger Name</label>
        <input id="passengerName" required>
      </div>
      <div>
        <label>Age</label>
        <input id="passengerAge" type="number" min="1" max="120" required>
      </div>
      <div>
        <label>Gender</label>
        <select id="passengerGender" required>
          <option value="">Select</option>
          <option>Male</option>
          <option>Female</option>
          <option>Other</option>
        </select>
      </div>
      <div>
        <label>Number of Tickets</label>
        <input id="ticketCount" type="number" min="1" max="6" value="1" required>
      </div>
      <div>
        <label>Payment Mode</label>
        <select id="paymentMode">
```

```
            <option>Online Demo Payment</option>
            <option>Offline / Counter Payment</option>
          </select>
        </div>
      </div>
      <p id="fareInfo" class="message"></p>
      <button type="submit">Confirm Booking</button>
      <button type="button" onclick="showSection('search')">Back</button>
    </form>
</section>

<!-- TICKET -->
<section id="ticket" class="panel hidden">
    <h2>🎉 Booking Successful!</h2>
    <div id="ticketDetails"></div>
    <br>
    <button onclick="window.print()">Print Ticket</button>
    <button onclick="showSection('history')">My Bookings</button>
</section>

<!-- PNR STATUS -->
<section id="pnr" class="panel hidden">
    <h2>Check PNR Status</h2>
    <label>Enter PNR Number</label>
    <input id="pnrInput" placeholder="Enter your PNR">
    <button onclick="checkPNR()">Check Status</button>
    <div id="pnrResult"></div>
</section>

<!-- CANCEL TICKET -->
<section id="cancel" class="panel hidden">
```

```
  <h2>Ticket Cancellation and Refund</h2>
  <label>Enter PNR Number</label>
  <input id="cancelPNR" placeholder="Enter PNR">
  <button class="danger" onclick="cancelTicket()">Cancel Ticket</button>
  <div id="cancelResult"></div>
  <p style="margin-top:15px">
    Demo refund policy: 90% refund if cancelled more than 2 days
    before the journey, 50% if cancelled within 2 days, and no
    refund after the journey date.
  </p>
</section>

<!-- BOOKING HISTORY -->
<section id="history" class="panel hidden">
  <h2>My Booking History</h2>
  <div id="bookingHistory"></div>
</section>

<!-- ADMIN -->
<section id="admin" class="panel hidden">
  <h2>Administrator Panel</h2>
  <p>Log in using the demo admin credentials to manage train records and reservations.</p>
  <br>
  <div id="adminLoginArea">
    <form onsubmit="adminLogin(event)">
      <label>Admin Email</label>
      <input id="adminEmail" type="email" required>
      <label>Admin Password</label>
      <input id="adminPassword" type="password" required>
      <button type="submit">Admin Login</button>
    </form>
```

```
    <div id="adminMsg"></div>
  </div>

  <div id="adminPanel" class="hidden">
    <h3>Add Train</h3>
    <form onsubmit="addTrain(event)">
      <div class="form-grid">
        <div><label>Train Number</label><input id="newNumber" required></div>
        <div><label>Train Name</label><input id="newName" required></div>
        <div><label>Origin</label><input id="newOrigin" required></div>
        <div><label>Destination</label><input id="newDestination" required></div>
        <div><label>Departure</label><input id="newDeparture" type="time" required></div>
        <div><label>Arrival</label><input id="newArrival" type="time" required></div>
        <div><label>Base Fare (₹)</label><input id="newFare" type="number" min="1"
required></div>
        <div><label>Available Seats</label><input id="newSeats" type="number" min="1"
required></div>
      </div>
      <button type="submit">Add Train</button>
    </form>

    <h3 style="margin-top:25px">Train Management</h3>
    <div id="adminTrains" class="table-wrap"></div>

    <h3 style="margin-top:25px">All Reservations and Payments</h3>
    <div id="adminBookings" class="table-wrap"></div>
  </div>
</section>

</div>

<footer>
```

```
  <p>© 2026 RailConnect | Online Train Reservation System</p>
  <p style="margin-top:5px">College Project Demonstration</p>
</footer>

<script>
// Demo data
const defaultTrains = [
  {id:101, number:"12601", name:"Chennai Express", origin:"Chennai", destination:"Bengaluru", departure:"06:00", arrival:"11:30", fare:450, seats:50},
  {id:102, number:"12602", name:"Bengaluru Express", origin:"Bengaluru", destination:"Chennai", departure:"07:30", arrival:"13:00", fare:450, seats:45},
  {id:103, number:"12603", name:"Kovai Express", origin:"Chennai", destination:"Coimbatore", departure:"07:15", arrival:"14:00", fare:550, seats:40},
  {id:104, number:"12604", name:"Pandian Express", origin:"Chennai", destination:"Madurai", departure:"21:00", arrival:"06:00", fare:600, seats:35},
  {id:105, number:"12605", name:"Deccan Express", origin:"Chennai", destination:"Hyderabad", departure:"18:00", arrival:"07:00", fare:700, seats:30},
  {id:106, number:"12606", name:"Mumbai Express", origin:"Chennai", destination:"Mumbai", departure:"20:00", arrival:"18:00", fare:900, seats:25}
];

function loadData(key, fallback) {
  try {
    const value = localStorage.getItem(key);
    return value ? JSON.parse(value) : fallback;
  } catch {
    return fallback;
  }
}

let trains = loadData("railTrains", defaultTrains);
let users = loadData("railUsers", []);
let bookings = loadData("railBookings", []);
```

```
let currentUser = null;
let isAdmin = false;
let selectedTrain = null;
let selectedClass = "Sleeper";

function saveData() {
  localStorage.setItem("railTrains", JSON.stringify(trains));
  localStorage.setItem("railUsers", JSON.stringify(users));
  localStorage.setItem("railBookings", JSON.stringify(bookings));
}

function showSection(id) {
  document.querySelectorAll(".container > section").forEach(section => {
    section.classList.add("hidden");
  });
  document.getElementById(id).classList.remove("hidden");

  if (id === "history") showHistory();
  if (id === "admin" && isAdmin) renderAdmin();
}

function registerUser(event) {
  event.preventDefault();

  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;

  if (users.some(u => u.email === email)) {
    document.getElementById("regMsg").innerHTML =
      '<p class="message">This email is already registered.</p>';
```

```
    return;
  }

  users.push({name, email, password});
  saveData();

  document.getElementById("regMsg").innerHTML =
    '<p class="message">Registration successful! Please log in.</p>';

  event.target.reset();
}

function loginUser(event) {
  event.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u => u.email === email && u.password === password);

  if (!user) {
    document.getElementById("loginMsg").innerHTML =
      '<p class="message">Invalid email or password. Please register first.</p>';
    return;
  }

  currentUser = user;
  document.getElementById("userLabel").textContent = "Welcome, " + user.name;
  document.getElementById("loginMsg").innerHTML =
    '<p class="message">Login successful!</p>';
```

```
  setTimeout(() => showSection("search"), 500);
}

function searchTrains(event) {
  event.preventDefault();

  if (!currentUser) {
    alert("Please log in before booking a ticket.");
    showSection("login");
    return;
  }

  const origin = document.getElementById("origin").value;
  const destination = document.getElementById("destination").value;
  const date = document.getElementById("journeyDate").value;
  selectedClass = document.getElementById("travelClass").value;

  if (origin === destination) {
    alert("Origin and destination must be different.");
    return;
  }

  if (date < today()) {
    alert("Please select today or a future journey date.");
    return;
  }

  const results = trains.filter(t =>
    t.origin.toLowerCase() === origin.toLowerCase() &&
    t.destination.toLowerCase() === destination.toLowerCase()
  );
```

```
  const resultBox = document.getElementById("trainResults");

  if (!results.length) {
    resultBox.innerHTML =
      '<p class="message">No direct trains found for this route in the demo.</p>';
    return;
  }

  resultBox.innerHTML = results.map(t => `
    <div class="train-card">
      <h3>${escapeHTML(t.name)} (${escapeHTML(t.number)})</h3>
      <div class="train-info">
        <span><b>${escapeHTML(t.origin)}</b> ${escapeHTML(t.departure)}</span>
        <span>→</span>
        <span><b>${escapeHTML(t.destination)}</b> ${escapeHTML(t.arrival)}</span>
      </div>
      <p>Class: ${escapeHTML(selectedClass)}</p>
      <p>Available seats: ${t.seats}</p>
      <p>Base fare: ₹${classFare(t.fare, selectedClass)} per passenger</p>
      <br>
      <button onclick="selectTrain(${t.id}, '${date}', '${selectedClass}')"
        ${t.seats < 1 ? "disabled" : ""}>Select Train</button>
    </div>
  `).join("");
}

function classFare(base, travelClass) {
  const multipliers = {
    "Sleeper": 1,
    "AC 3 Tier": 1.7,
```

```
    "AC 2 Tier": 2.4,
    "Chair Car": 1.3
  };
  return Math.round(base * multipliers[travelClass]);
}

function selectTrain(id, date, travelClass) {
  if (!currentUser) {
    alert("Please log in first.");
    showSection("login");
    return;
  }

  selectedTrain = trains.find(t => t.id === id);
  selectedClass = travelClass;

  if (!selectedTrain || selectedTrain.seats < 1) {
    alert("No seats are available.");
    return;
  }

  document.getElementById("selectedTrain").textContent =
    selectedTrain.name + " | " + selectedTrain.origin + " to " +
    selectedTrain.destination + " | " + date + " | " + travelClass;

  document.getElementById("fareInfo").textContent =
    "Fare per passenger: ₹" +
    classFare(selectedTrain.fare, selectedClass) +
    ". Final fare depends on the number of passengers.";

  document.getElementById("booking").dataset.date = date;
```

```
  document.getElementById("ticketCount").max = Math.min(6, selectedTrain.seats);
  document.getElementById("ticketCount").value = 1;

  showSection("booking");
}

function confirmBooking(event) {
  event.preventDefault();

  if (!currentUser || !selectedTrain) {
    alert("Please log in and select a train.");
    return;
  }

  const count = Number(document.getElementById("ticketCount").value);

  // Recheck current availability before booking.
  const train = trains.find(t => t.id === selectedTrain.id);

  if (!train || count < 1 || count > 6 || count > train.seats) {
    alert("The requested number of seats is not available.");
    return;
  }

  const date = document.getElementById("booking").dataset.date;

  if (date < today()) {
    alert("Journey date has passed.");
    return;
  }
```

```
const fare = classFare(train.fare, selectedClass) * count;
const mode = document.getElementById("paymentMode").value;

// Offline payments are pending until an actual payment is collected.
const paymentStatus = mode.startsWith("Offline")
  ? "Pending - Counter Payment"
  : "Demo Payment Successful";

const booking = {
  pnr: String(Date.now()).slice(-10),
  userEmail: currentUser.email,
  passenger: document.getElementById("passengerName").value.trim(),
  age: Number(document.getElementById("passengerAge").value),
  gender: document.getElementById("passengerGender").value,
  trainId: train.id,
  trainName: train.name,
  trainNumber: train.number,
  origin: train.origin,
  destination: train.destination,
  date,
  travelClass: selectedClass,
  count,
  fare,
  paymentMode: mode,
  paymentStatus,
  status: "CONFIRMED",
  created: new Date().toLocaleString()
};

if (mode.startsWith("Offline")) {
  // Reserve seats, but do not falsely claim payment has been completed.
```

```
      booking.status = "AWAITING PAYMENT";
    }

    train.seats -= count;
    bookings.push(booking);
    saveData();

    document.getElementById("ticketDetails").innerHTML = `
      <div class="message">
        <h3>Reservation Details</h3>
        <p><b>PNR:</b> ${booking.pnr}</p>
        <p><b>Passenger:</b> ${escapeHTML(booking.passenger)}</p>
        <p><b>Train:</b> ${escapeHTML(booking.trainName)}
(${escapeHTML(booking.trainNumber)})</p>
        <p><b>From:</b> ${escapeHTML(booking.origin)}</p>
        <p><b>To:</b> ${escapeHTML(booking.destination)}</p>
        <p><b>Journey Date:</b> ${escapeHTML(booking.date)}</p>
        <p><b>Class:</b> ${escapeHTML(booking.travelClass)}</p>
        <p><b>Tickets:</b> ${booking.count}</p>
        <p><b>Total Fare:</b> ₹${booking.fare}</p>
        <p><b>Reservation Status:</b> ${escapeHTML(booking.status)}</p>
        <p><b>Payment Status:</b> ${escapeHTML(booking.paymentStatus)}</p>
      </div>
    `;

    showSection("ticket");
}

function checkPNR() {
    const pnr = document.getElementById("pnrInput").value.trim();
    const booking = bookings.find(b => b.pnr === pnr);
```

```
  const box = document.getElementById("pnrResult");

  if (!booking) {
    box.innerHTML = '<p class="message">PNR not found.</p>';
    return;
  }

  box.innerHTML = `
    <div class="message">
      <h3>PNR Details</h3>
      <p>Train: ${escapeHTML(booking.trainName)}</p>
      <p>Passenger: ${escapeHTML(booking.passenger)}</p>
      <p>Journey: ${escapeHTML(booking.origin)} → ${escapeHTML(booking.destination)}</p>
      <p>Date: ${escapeHTML(booking.date)}</p>
      <p>Class: ${escapeHTML(booking.travelClass)}</p>
      <p>PNR Status: <b>${escapeHTML(booking.status)}</b></p>
      <p>Payment Status: ${escapeHTML(booking.paymentStatus)}</p>
    </div>
  `;
}

function cancelTicket() {
  const pnr = document.getElementById("cancelPNR").value.trim();
  const booking = bookings.find(b => b.pnr === pnr);
  const box = document.getElementById("cancelResult");

  if (!booking) {
    box.innerHTML = '<p class="message">PNR not found.</p>';
    return;
  }
```

```
  if (currentUser && !isAdmin && booking.userEmail !== currentUser.email) {
    box.innerHTML = '<p class="message">You cannot cancel another user's ticket.</p>';
    return;
  }

  if (!currentUser && !isAdmin) {
    box.innerHTML = '<p class="message">Please log in before cancellation.</p>';
    showSection("login");
    return;
  }

  if (booking.status === "CANCELLED") {
    box.innerHTML = '<p class="message">This ticket is already cancelled.</p>';
    return;
  }

  if (booking.status === "AWAITING PAYMENT") {
    box.innerHTML = '<p class="message">This booking has pending payment. Contact the counter to cancel it.</p>';
    return;
  }

  if (booking.status !== "CONFIRMED") {
    box.innerHTML = '<p class="message">This ticket cannot be cancelled.</p>';
    return;
  }

  if (booking.date < today()) {
    box.innerHTML = '<p class="message">Journey date has passed. Refund is unavailable in this demo.</p>';
    return;
  }
```

```
const daysLeft = Math.ceil(
   (new Date(booking.date + "T00:00:00") -
    new Date(today() + "T00:00:00")) / 86400000
);

const percentage = daysLeft > 2 ? 90 : 50;
const refund = Math.round(booking.fare * percentage / 100);

if (!confirm("Cancel this ticket? Estimated refund: ₹" + refund +
   ". The refund is a demo calculation.")) return;

booking.status = "CANCELLED";
booking.refund = refund;
booking.refundStatus = "Pending admin processing";

const train = trains.find(t => t.id === booking.trainId);
if (train) train.seats += booking.count;

saveData();

box.innerHTML = `
   <div class="message">
      <h3>Cancellation Successful</h3>
      <p>PNR: ${booking.pnr}</p>
      <p>Original Fare: ₹${booking.fare}</p>
      <p>Demo Refund Rate: ${percentage}%</p>
      <p>Estimated Refund: ₹${refund}</p>
      <p>Refund Status: Pending admin processing</p>
   </div>
`;
```

```
}

function showHistory() {
  const box = document.getElementById("bookingHistory");

  if (!currentUser) {
    box.innerHTML = '<p class="message">Please log in to view your bookings.</p>';
    return;
  }

  const list = bookings.filter(b => b.userEmail === currentUser.email);

  if (!list.length) {
    box.innerHTML = "<p>No bookings found.</p>";
    return;
  }

  box.innerHTML = `<div class="table-wrap"><table>
    <tr><th>PNR</th><th>Train</th><th>Journey Date</th>
    <th>Fare</th><th>Status</th><th>Action</th></tr>
    ${list.map(b => `
    <tr>
      <td>${b.pnr}</td>
      <td>${escapeHTML(b.trainName)}</td>
      <td>${escapeHTML(b.date)}</td>
      <td>₹${b.fare}</td>
      <td>${escapeHTML(b.status)}</td>
      <td><button onclick="printPNR('${b.pnr}')">Print</button></td>
    </tr>`).join("")}
  </table></div>`;
}
```

```
function printPNR(pnr) {
  const b = bookings.find(x => x.pnr === pnr);

  if (!b || (!isAdmin && (!currentUser || b.userEmail !== currentUser.email))) {
    alert("Booking not found or access denied.");
    return;
  }

  document.getElementById("ticketDetails").innerHTML = `
    <h3>RailConnect E-Ticket</h3>
    <p>PNR: ${b.pnr}</p>
    <p>Passenger: ${escapeHTML(b.passenger)}</p>
    <p>Train: ${escapeHTML(b.trainName)} (${escapeHTML(b.trainNumber)})</p>
    <p>From: ${escapeHTML(b.origin)} → ${escapeHTML(b.destination)}</p>
    <p>Date: ${escapeHTML(b.date)}</p>
    <p>Class: ${escapeHTML(b.travelClass)}</p>
    <p>Passengers: ${b.count}</p>
    <p>Fare: ₹${b.fare}</p>
    <p>Status: ${escapeHTML(b.status)}</p>
  `;

  showSection("ticket");
}

function adminLogin(event) {
  event.preventDefault();

  const email = document.getElementById("adminEmail").value;
  const password = document.getElementById("adminPassword").value;
```

```
  if (email === "admin@rail.com" && password === "admin123") {
    isAdmin = true;
    document.getElementById("adminLoginArea").classList.add("hidden");
    document.getElementById("adminPanel").classList.remove("hidden");
    renderAdmin();
  } else {
    document.getElementById("adminMsg").innerHTML =
      '<p class="message">Invalid administrator credentials.</p>';
  }
}

function addTrain(event) {
  event.preventDefault();

  trains.push({
    id: Date.now(),
    number: document.getElementById("newNumber").value.trim(),
    name: document.getElementById("newName").value.trim(),
    origin: document.getElementById("newOrigin").value.trim(),
    destination: document.getElementById("newDestination").value.trim(),
    departure: document.getElementById("newDeparture").value,
    arrival: document.getElementById("newArrival").value,
    fare: Number(document.getElementById("newFare").value),
    seats: Number(document.getElementById("newSeats").value)
  });

  saveData();
  event.target.reset();
  renderAdmin();
  alert("Train added successfully.");
}
```

```
function renderAdmin() {
  if (!isAdmin) return;

  document.getElementById("adminTrains").innerHTML = `
    <table>
    <tr><th>Number</th><th>Name</th><th>Route</th><th>Fare</th>
    <th>Seats</th><th>Action</th></tr>
    ${trains.map(t => `
      <tr>
        <td>${escapeHTML(t.number)}</td>
        <td>${escapeHTML(t.name)}</td>
        <td>${escapeHTML(t.origin)} → ${escapeHTML(t.destination)}</td>
        <td>₹${t.fare}</td>
        <td>${t.seats}</td>
        <td><button class="danger" onclick="deleteTrain(${t.id})">Delete</button></td>
      </tr>
    `).join("")}
    </table>`;

  document.getElementById("adminBookings").innerHTML = `
    <table>
    <tr><th>PNR</th><th>Passenger</th><th>Train</th><th>Fare</th>
    <th>Payment</th><th>Status</th><th>Refund</th></tr>
    ${bookings.map(b => `
      <tr>
        <td>${b.pnr}</td>
        <td>${escapeHTML(b.passenger)}</td>
        <td>${escapeHTML(b.trainName)}</td>
        <td>₹${b.fare}</td>
        <td>${escapeHTML(b.paymentStatus)}</td>
```

```
            <td>${escapeHTML(b.status)}</td>
            <td>
              ${b.refund !== undefined
                ? "₹" + b.refund + " - " + escapeHTML(b.refundStatus || "")
                : "-"}
              ${b.refund !== undefined && b.refundStatus !== "Processed"
                ? `<button class="green" onclick="processRefund('${b.pnr}')">Process</button>`
                : ""}
            </td>
          </tr>
        `).join("")}
      </table>`;
}

function processRefund(pnr) {
  if (!isAdmin) return;

  const booking = bookings.find(b => b.pnr === pnr);

  if (!booking || booking.status !== "CANCELLED" ||
    booking.refundStatus === "Processed") return;

  booking.refundStatus = "Processed";
  saveData();
  renderAdmin();

  alert("Demo refund marked as processed. No actual money was transferred.");
}

function deleteTrain(id) {
  if (!isAdmin) return;
```

```
  const hasActiveBookings = bookings.some(b =>
    b.trainId === id && b.status !== "CANCELLED"
  );

  if (hasActiveBookings) {
    alert("Cannot delete a train with active reservations.");
    return;
  }

  if (!confirm("Delete this train?")) return;

  trains = trains.filter(t => t.id !== id);
  saveData();
  renderAdmin();
}

function today() {
  const d = new Date();
  const offset = d.getTimezoneOffset();
  return new Date(d.getTime() - offset * 60000).toISOString().slice(0, 10);
}

function escapeHTML(value) {
  return String(value ?? "").replace(/[&<>"']/g, ch => ({
    "&": "&amp;",
    "<": "&lt;",
    ">": "&gt;",
    '"': "&quot;",
    "'": "&#39;"
  })[ch]);
```

```
}

// Set minimum journey date.
document.getElementById("journeyDate").min = today();
document.getElementById("journeyDate").value = today();

// Show the home page initially.
showSection("home");
</script>

</body>
</html>
```